Tool Radius Compensation

Digital Manufacturing



Paolo Bosetti

Updated: 14/10/2026

Table of contents

1 Tool radius compensation (TRC) 1
1.1 TRC and G-code . . . 2
1.2 Tool radius compensation . . . 2
1.3 Tool radius compensation: Line-Line . . . 3
1.4 Tool radius compensation: Line-Circle . . . 3
1.5 Line-Line case . . . 3
1.6 Line-Line case . . . 3
1.7 Line-Line case . . . 4
1.8 Line-Line case . . . 5
1.9 Line-Arc (or Arc-Line) . . . 5
1.10 Line-Arc (or Arc-Line) . . . 6
1.11 Code changes . . . 7
1.12 Code changes . . . 7
1.13 Code changes . . . 8
1.14 Code changes . . . 8
1.15 Advanced TRC . . . 8
1.16 Rules . . . 8

1 Tool radius compensation (TRC)

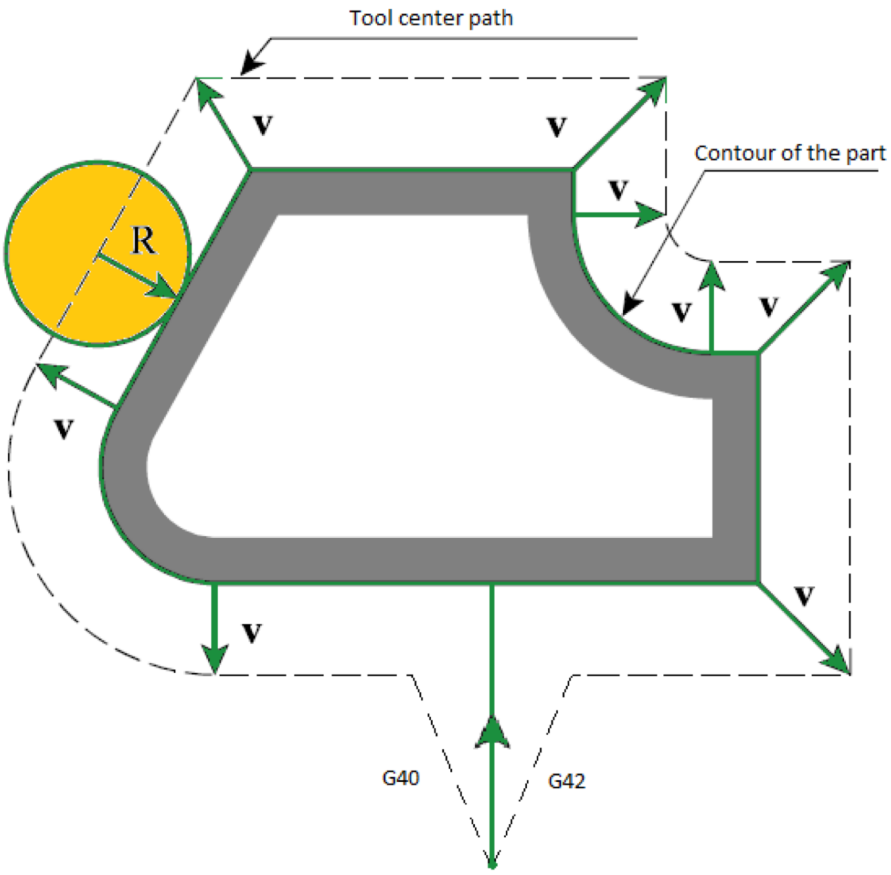
A possible feature to implement for the personal project

1.1 TRC and G-code

- G-code standard supports a feature for offsetting the tool trajectory to the left or to the right of the trajectory by an amount equal to the tool radius
- This is only affecting the X-Y projection of the trajectory, Z values remain unchanged
- The TRC is set to the left with `G41`, to the right with `G42` and is disabled with `G40`
- TRC is fully active at the end of the current block

1.2 Tool radius compensation

- TRC offsets the tool center path (programmed points) according to `G41/G42`
- New target points are calculated for each block on the basis of the offset intersection with the next block (arc or line)
- Disabled for rapid motions



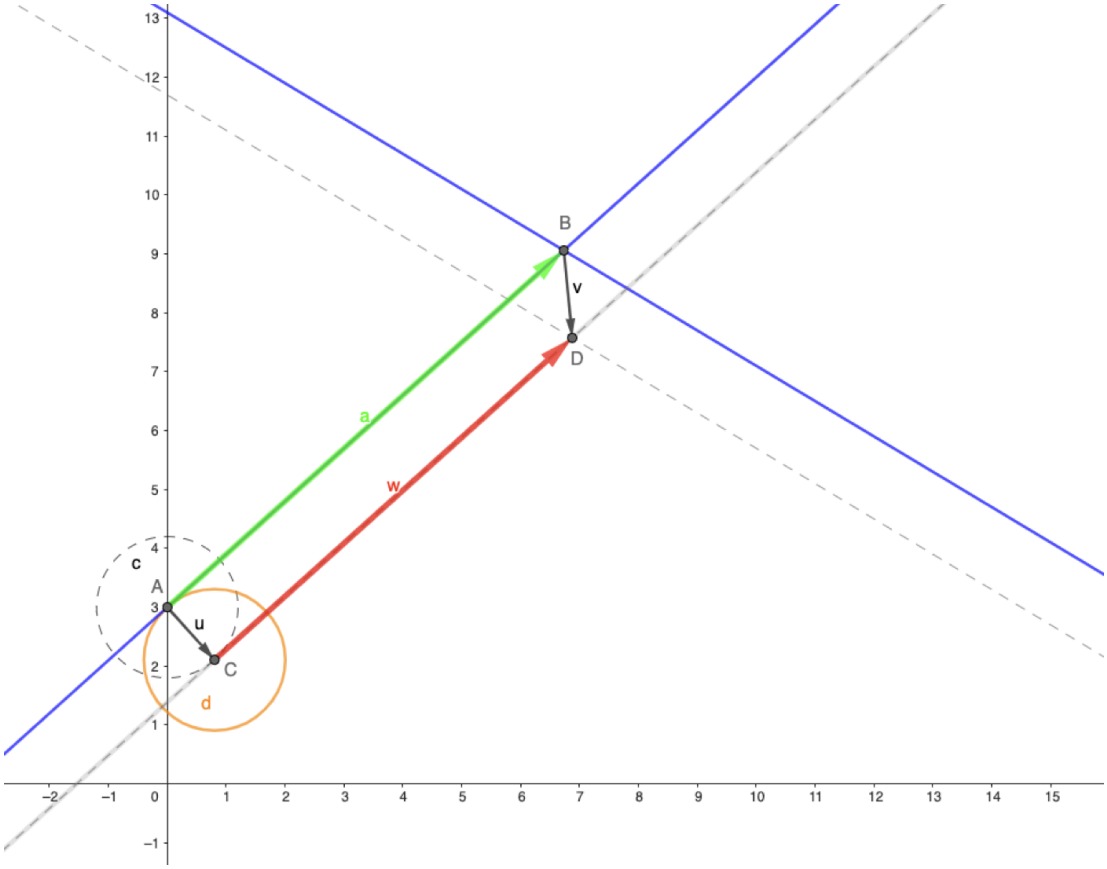
1.3 Tool radius compensation: Line-Line

1.4 Tool radius compensation: Line-Circle

1.5 Line-Line case

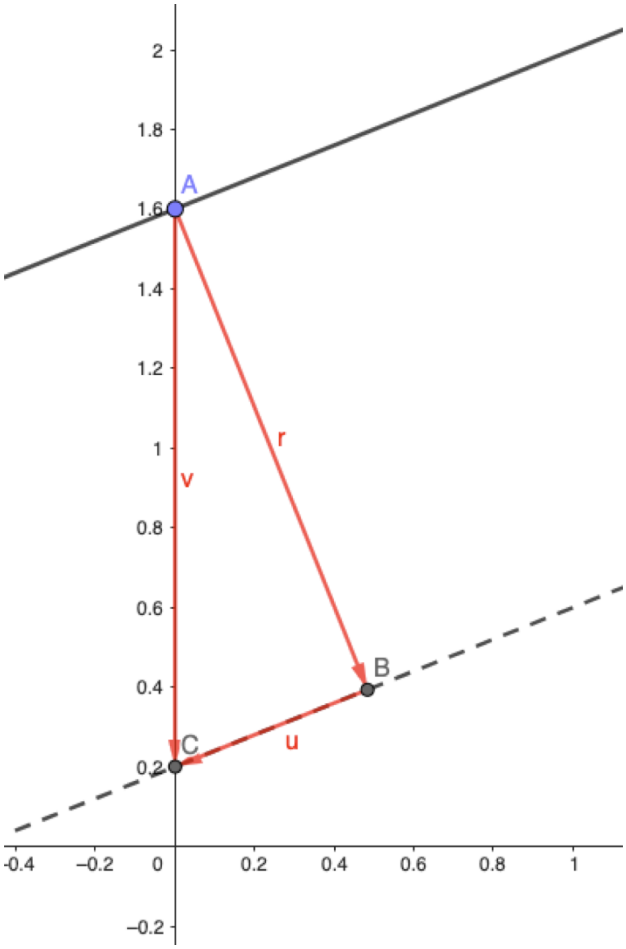
We need to calculate the point D:

1. Given the line equation $y = ax + b$, we need to calculate the equation of the offset line(s)
2. Calculate D as intercept between the two offset lines



1.6 Line-Line case

- Given $y = ax + b$, the equation of the offset line is $y = ax + b - v$
- Being $a = u/r$, it follows that: $v = \sqrt{u^2 + r^2} = r\sqrt{a^2 + 1}$
- So, $y = ax + b \pm r\sqrt{a^2 + 1}$
- **The sign depends on G41/G42 and on the path direction**



1.7 Line-Line case

Intersection D between

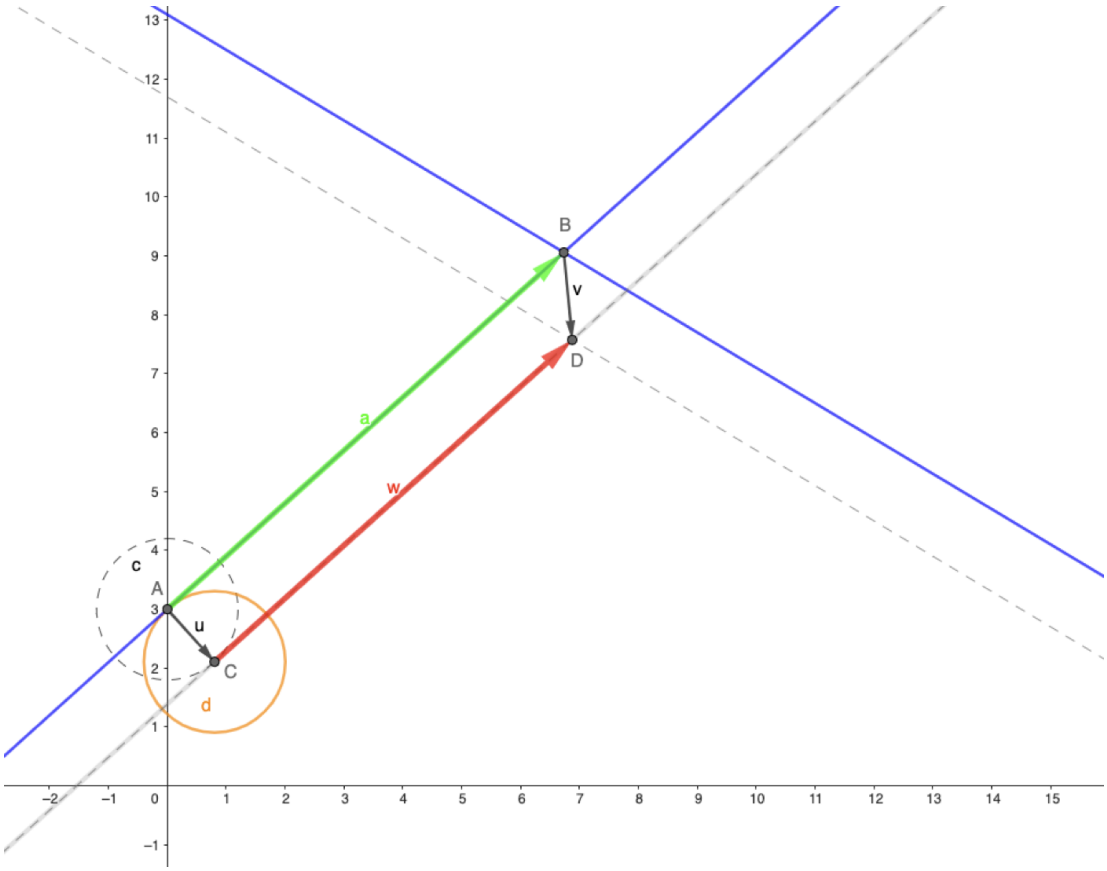
$y = a_1 x + b_1 - v_1 = a_1 x + b_{o,1}$

and

$y = a_2 x + b_2 - v_2 = a_2 x + b_{o,2}$

has coordinates:

$$(x_i, y_i) = \left(\frac{b_{o,2} - b_{o,1}}{a_1 - a_2}, \frac{b_{o,1} a_2 - b_{o,2} a_1}{a_2 - a_1} \right)$$



1.8 Line-Line case

- **Exception:** when two segments are collinear ($a = a_1 = a_2$), the intercept is undefined
- The new intersection node is set at an offset r **perpendicular** to the segments

$$C = \begin{cases} (0, -r), & \text{when } a = 0 \\ r(s, -s/a) \text{ with } s = \frac{a}{|a|}\sqrt{\frac{a^2}{1+a^2}}, & \text{otherwise} \end{cases}$$

1.9 Line-Arc (or Arc-Line)

Similar to line-line case, but with a caveat:

- Line-circle intersection gives two points, G and B_r: you have to select the intersection that is closer to the node to be offset (B in figure)

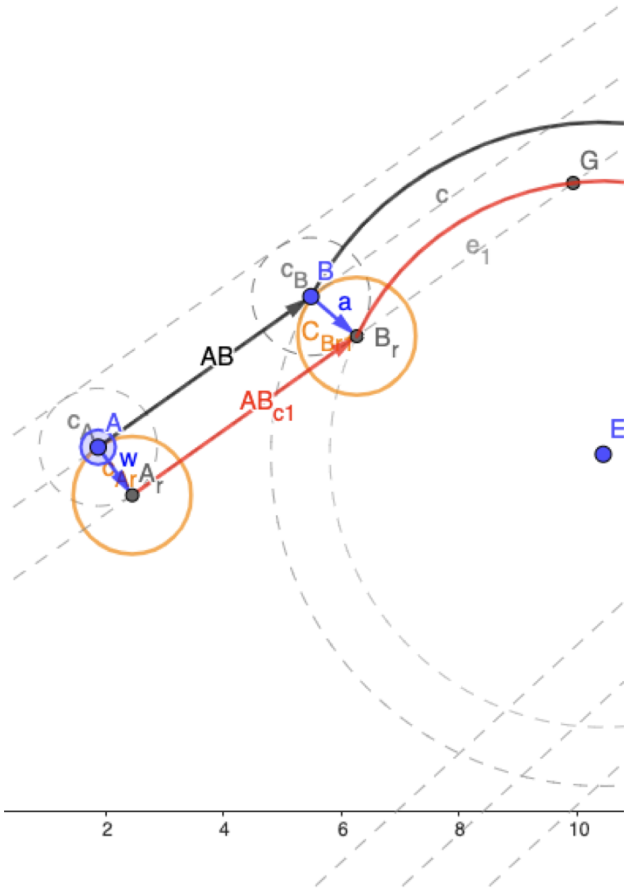


Figure 1: Non-tangent line-arc case

1.10 Line-Arc (or Arc-Line)

Similar to line-line case, but with a caveat:

- Line-circle intersection gives two points, G and B_r: you have to select the intersection that is closer to the node to be offset (B in figure)

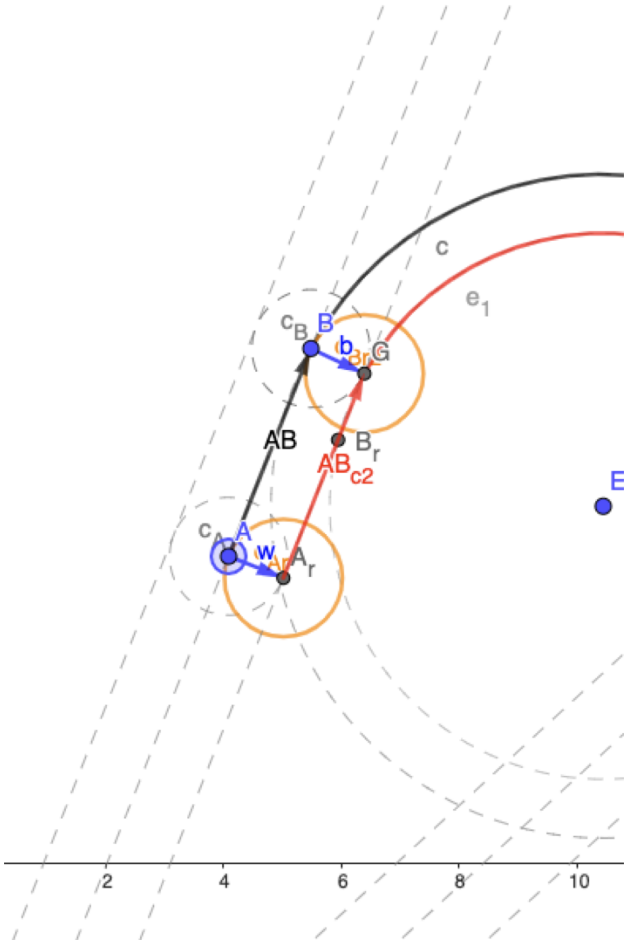


Figure 2: Tangent line-arc case

1.11 Code changes

Support for G40/G41/G42

- An `enum class TRCType` with `NONE`, `LEFT`, `RIGHT`
- An `TRCType _trc` field in the `BlockTRC` class
- Proper code in `BlockTRC::parse()`
- `_trc` shall be managed as a *modal* field (i.e. inherited from previous block)

1.12 Code changes

Support for tool radius

- The YAML file needs to have a field as: `tools: [10.0, 8.0, 5.0, 2.5]` giving the list of available tools and respective radiuses
- The `Machine` class needs to properly load the tool list into a new field: `vector<data_t> _tools`
- Add `machine::tool_radius()`, which returns the radius of a given tool number

1.13 Code changes

Support for offset calculation

- The method `Block::parse()` needs to change the target of the previous block if the previous block has `_trc != TRCType::NONE`
- The target must be changed according to the equations shown above and the radius of the current tool

1.14 Code changes

Support for offset calculation

- Equations for intersections possibly require:
  - `Block::angle_with_prev()`: angle in radians with previous block
  - `Block::equation()`: returns a and b as output parameters (refs); for an arc, parameters of the implicit equation $(x - x_c)^2 + (y - y_c)^2 = r^2$ are available already
- These functions can be private

1.15 Advanced TRC

1.16 Rules

- You must work alone (I'll check and compare results!)
- Document progress with commits published on GitHub (**private** repo shared with me)
- No help from my side, except these slides
- When you are done, you must discuss your work with me
  - If it works (no warnings, errors/errors, works as expected) and it is not copied, you get 30/30
- If you implement advanced TRC or impress me with some other clever addition, you get 30/30 *cum laude*
- Must be done before the end of next year